Tempered Langevin diffusions with heavy-tailed targets: explosions, moment thresholds and ergodicity

Aleksandar Mijatović and Haokai Xie
Department of Statistics, University of Warwick

Part 1: Main Results

TLDs with polynomially decaying targets

Let Π be a probability measure on $\mathbb{R}^d$ and $\Pi(\mathrm{d}x)/\mathrm{d}x \propto \pi(x)$ where $\pi \in C^2(\mathbb{R}^d)$ satisfies

$$|x|^{d+r}\pi(x) \to c_\pi > 0 \quad \text{and} \quad \langle x, \nabla \log \pi(x)\rangle \to -(d+r) \quad \text{as} \quad |x| \to \infty^1$$

for some $r > 0$. The *tempered Langevin diffusion* X with temperature $\ell \geq 0$ solves the SDE

$$\mathrm{d}X_t = (1-2\ell)\pi(X_t)^{-2\ell}\nabla \log \pi(X_t)\mathrm{d}t + \sqrt{2}\pi(X_t)^{-\ell}\,\mathrm{I}_d\,\mathrm{d}B_t$$

with stationary measure Π. As shown in [FR05; BM26], we have



Question: which property of X_t drives this transition in convergence rates?

[1] For $f : \mathbb{R}^d \to \mathbb{R}$, we write $f(x) \to c \in \mathbb{R}$ as $|x| \to \infty$ if $\lim_{\rho\to\infty} \sup_{|x|=\rho} |f(x) - c| = 0$.

Main results: explosions, moment thresholds and ergodicity

Assumption: $|x|^{d+r}\pi(x) \to c_\pi > 0$ and $\langle x, \nabla \log \pi(x)\rangle \to -(d+r)$ as $|x| \to \infty$.

	Explosion	**Ergodicity in f_p-variation**
$0 \le \ell < \frac{1}{d+r}$	$\zeta = \infty$ a.s.	$\approx t^{-\frac{r-p}{2(1-\ell(d+r))}}\ \forall\, p \in (0, r)$
$\ell = \frac{1}{d+r}$	$\zeta = \infty$ a.s.	Exponential $\forall\, p \in (0, r)$,
$\frac{1}{d+r} < \ell < \frac{1+r/2}{d+r}$	$\zeta = \infty$ a.s.	Uniform for $p = 0$
$\ell = \frac{1+r/2}{d+r}$		
$\frac{1+r/2}{d+r} < \ell < \infty$		

Main results: explosions, moment thresholds and ergodicity

Assumption: $|x|^{d+r}\pi(x) \to c_\pi > 0$ and $\langle x, \nabla \log \pi(x)\rangle \to -(d+r)$ as $|x| \to \infty$.

	Explosion	**Moment thresholds of $\lvert X_t\rvert^p$ $(p>0)$**	**Ergodicity in f_p-variation & L^p-Wasserstein**
$0 \le \ell < \frac{1}{d+r}$	$\zeta = \infty$ a.s.	$\forall t>0 : \mathbb{E}_x[e^{\lvert X_t\rvert^p}] < \infty \iff p \in (0, 2(1-\ell(d+r)))$	$\approx t^{-\frac{r-p}{2(1-\ell(d+r))}}\ \forall\, p \in (0,r)$
$\ell = \frac{1}{d+r}$	$\zeta = \infty$ a.s.	$\forall t>0 : \mathbb{E}_x[\lvert X_t\rvert^p] < \infty$, but $\mathbb{E}_x[e^{\lvert X_t\rvert^p}] = \infty$	Exponential $\forall\, p \in (0,r)$, but not uniform
$\frac{1}{d+r} < \ell < \frac{1+r/2}{d+r}$	$\zeta = \infty$ a.s.	$\mathbb{E}_x[\lvert X_t\rvert^p] < \infty \iff p \in (0,r)$	Uniform $\forall\, p \in (0,r)$
$\ell = \frac{1+r/2}{d+r}$	$\zeta = \infty$ a.s.	$\mathbb{E}_x[\lvert X_t\rvert^p] < \infty \iff p \in (0,r)$	Uniform $\forall\, p \in (0,r)$
$\frac{1+r/2}{d+r} < \ell < \infty$	$\zeta < \infty$ a.s.	\	\

Ergodicity decay rates: $\|\mathbb{P}_x(X_t \in \cdot) - \Pi(\cdot)\|_{f_p}$, for $f_p(x) = 1 + |x|^p$, and $\mathcal{W}_p(\mathbb{P}_x(X_t \in \cdot), \Pi)^{p \vee 1}$, for $\boldsymbol{p > 0}$ (since $r > 0$), where $\mathcal{W}_p(\mu,\nu) = \inf_{\eta \in C(\mu,\nu)} (\int_{\mathbb{R}^d \times \mathbb{R}^d} |x-y|^p \, \eta(\mathrm{d}x, \mathrm{d}y))^{1/(p\vee 1)}$.

Intuition: critical temperature thresholds

Assume $\pi(x) = |x|^{-d-r}$ for $|x|$ sufficiently large and let $\mathcal{A}$ be the *extended generator* of X.

$$\mathcal{A}|x|^\gamma = -\gamma((r-2\alpha)-\gamma)(|x|^\gamma)^{1+\frac{2\alpha}{\gamma}}, \quad \text{where} \quad \alpha := \ell(d+r)-1,$$

for $|x|$ large enough. Both critical temperature thresholds can be read off.

	$-\gamma((r-2\alpha)-\gamma)$	$u \mapsto u^{1+\frac{2\alpha}{\gamma}}$
$0 \le \ell < \frac{1}{d+r}$	< 0, if $0 < \gamma < 2\alpha$	sublinear ($\alpha < 0$)
$\ell = \frac{1}{d+r}$	< 0, if $0 < \gamma < 2\alpha$	linear ($\alpha = 0$)
$\frac{1}{d+r} < \ell < \frac{1+r/2}{d+r}$	< 0, if $0 < \gamma < 2\alpha$	superlinear ($\alpha > 0$)
$\ell = \frac{1+r/2}{d+r}$	$= \gamma^2$	superlinear ($\alpha > 0$)
$\frac{1+r/2}{d+r} < \ell < \infty$	> 0, for all $\gamma > 0$	superlinear ($\alpha > 0$)

Note that at $\ell = \frac{1+r/2}{d+r}$, $\mathcal{A}\log|x| = (2\alpha - r)|x|^{2\alpha} = 0$ for all sufficiently large $|x|$.

Intuition: moments properties

Assume $\pi(x) = |x|^{-d-r}$ whenever $|x| \geq R$ for some $R > 0$. Recall that $\alpha = \ell(d+r) - 1$. Consider a C^2-function $\Phi : \mathbb{R}^d \to (0, \infty)$ s.t. for $|x| \geq R$,

$$\Phi(x) = \begin{cases} |x|^{-\alpha}/\big(\sqrt{2}|\alpha|\big), & \text{if} \quad \ell \neq \frac{1}{d+r}, \\ \log|x|/\sqrt{2}, & \text{if} \quad \ell = \frac{1}{d+r}. \end{cases}$$

Let $|x_0| > R$ and $\tau_R = \inf\{t > 0 : |X_t| \leq R\}$. Itô's formula gives that, on $\{t \leq \tau_R\}$,

$$\Phi(X_t) = \begin{cases} \Phi(x_0) - \frac{r+|\alpha|}{\sqrt{2}} \int_0^t |X_s|^\alpha \mathrm{d}s + W_t, & 0 \leq \ell \leq \frac{1}{d+r} \iff \alpha \in [-1, 0], \\ \Phi(x_0) + \int_0^t \frac{\delta_\star - 1}{2\Phi(X_s)} \mathrm{d}s + W_t, & \frac{1}{d+r} < \ell < \infty \iff \alpha > 0, \end{cases}$$

where W is a standard Brownian motion on $\mathbb{R}$ and $\delta_\star := \frac{r}{\alpha} > 0$ when $\ell > \frac{1}{d+r}$.

Intuitions behind moments properties (continued)

Let Z be a Bessel process of dimension $\delta_\star$ with $Z_0 = z \geq 0$ and let $t > 0$.

- **Finite explosion time when** $\ell > \frac{1+r/2}{d+r} \iff \alpha = \ell(d+r) - 1 > \frac{r}{2}$:

$$Z \text{ hits zero a.s.} \iff \delta_\star = \frac{r}{\alpha} \in (0,2) \longleftrightarrow \Phi(X) \text{ “reaches” } \infty \text{ a.s.} \iff \ell > \frac{1+r/2}{d+r}.$$

- **Finite p-th moments when** $\ell \in \left(\frac{1}{d+r}, \frac{1+r/2}{d+r}\right] \iff \alpha = \ell(d+r) - 1 \in \left(0, \frac{r}{2}\right]$:

$$\mathbb{E}\left[Z_t^{-\nu}\right] < \infty \iff \nu = \frac{p}{\alpha} \in [0, \delta_\star) \longleftrightarrow \mathbb{E}[|X_t|^p] \approx \mathbb{E}\left[\Phi(X_t)^{-\nu}\right] < \infty \iff p \in [0, r).$$

Moreover, these upper bounds on p-th moments are **uniform in space**, because

$$\mathbb{E}_0\left[Z_t^{-\nu}\right] < \infty \text{ for } \nu \in [0, \delta_\star) \longleftrightarrow \mathbb{E}_\infty[|X_t|^p] < \infty \text{ for } p \in [0, r).$$

References

[BM26] Miha Brešar and Aleksandar Mijatović. “Subexponential lower bounds for f-ergodic Markov processes”. In: *Probability Theory and Related Fields* 194 (2026), pp. 997–1054.

[FR05] Gersende Fort and Gareth Roberts. “Subgeometric Ergodicity of Strong Markov Processes”. In: *The Annals of Applied Probability* 15.2 (2005), pp. 1565–1589.